REACHING A BOUNDARY POINT BY A DVR VALUED IN THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2799-2813 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2815-2838, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 39 of queue_stacks-stacks-more-morphisms.json (pb=1156, stmt 2799-2813, proof 2815-2838).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-reach-point-closure-Noetherian' also exists in stacks-limits, stacks-spaces-limits — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0CM2: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2838–2838. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04XH: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2838–2838. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Assume f finite type and $\mathcal{Y}$ locally Noetherian. Let $y \in |\mathcal{Y}|$ be a point in the closure of the image of $|f|$. Then there exists a commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & \mathcal{Y} \end{array}$$

of algebraic stacks where A is a discrete valuation ring and K is its field of fractions mapping the closed point of $\mathrm{Spec}(A)$ to y.

Proof. Choose an affine scheme V, a point $v \in V$ and a smooth morphism $V \to \mathcal{Y}$ mapping v to y. The map $|V| \to |\mathcal{Y}|$ is open and by Properties of Stacks, Lemma 04XH the image of $|\mathcal{X} \times_{\mathcal{Y}} V| \to |V|$ is the inverse image of the image of $|f|$. We conclude that the point v is in the closure of the image of $|\mathcal{X} \times_{\mathcal{Y}} V| \to |V|$. If we prove the lemma for $\mathcal{X} \times_{\mathcal{Y}} V \to V$ and the point v, then the lemma follows for f and y. In this way we reduce to the situation described in the next paragraph.

Assume we have $f : \mathcal{X} \to Y$ and $y \in |Y|$ as in the lemma where Y is a Noetherian affine scheme. Since f is quasi-compact, we conclude that $\mathcal{X}$ is quasi-compact. Hence we can choose an affine scheme W and a surjective smooth morphism $W \to \mathcal{X}$. Then the image of $|f|$ is the same as the image of $|W| \to |Y|$. In this way we reduce to the case of schemes which is Limits, Lemma 0CM2. □